

Abstract

Paper IV of this series derived the master self-consistency equation for the CDFT vacuum density ρ_0 and identified the vacuum chirality phase ϕ_c as the single remaining unknown. The present paper derives ϕ_c from the $\mathbb{Z}_3$ algebraic structure of the trefoil vortex itself, closing the loop.

The argument is a self-consistency condition: the Brannen $\mathbb{Z}_3$ parametrisation of the lepton masses has a universal invariant $Q = 2/3$ (the Koide ratio, proved in Paper II). The chirality equilibrium condition (Paper III) requires $3\theta + \phi_c = \pi$. Self-consistency demands that the chirality breaking angle 3θ equals the $\mathbb{Z}_3$ invariant Q :

$$3\theta = Q = \frac{2}{3}, \quad \Rightarrow \quad \phi_c = \pi - \frac{2}{3}.$$

This gives $\theta = 2/9$ rad exactly.

With one input (m_e as the mass unit), the theory predicts:

$$\begin{aligned} m_\mu &= 105.659 \text{ MeV} \quad (+9.8 \text{ ppm from PDG 2022}) \\ m_\tau &= 1776.985 \text{ MeV} \quad (+0.125 \text{ MeV} = +1.04 \sigma \text{ from PDG 2022}) \end{aligned}$$

The Koide formula $Q = 2/3$ holds exactly within the parametrisation. The CDFT lepton-sector construction is therefore internally closed, with m_e as the external mass-unit input and no additional fitted parameter in this step.

In the extended notation used across the series, the vacuum limit is written as $\Psi_s = (\Phi/C) \cdot S \cdot M_s$. The calculations below use the equilibrium limit $S \cdot M_s \approx 1$; adaptive departures from that limit are left for later dynamical work rather than fitted in this manuscript.

Zero Parameters: Deriving the Vacuum Chirality Phase from the Koide Invariant

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1 Introduction

The Constraint-Driven Field Theory (CDFT) series has, paper by paper, derived the structure of the lepton sector from the geometry of the vacuum medium:

1. **Paper I** [1]: $\alpha = 1/137.036$ from vortex ring equilibrium.
2. **Paper II** [2]: Koide formula $Q = 2/3$ as a theorem of $\mathbb{Z}_3$ symmetry; leptons as trefoil $T(2,3)$ modes.
3. **Paper III** [3]: All four open problems closed; ρ_0 identified as the sole remaining unknown. Chirality equilibrium: $3\theta + \phi_c = \pi$.
4. **Paper IV** [4]: Master self-consistency equation for ρ_0 ; Compton–ring coincidence; $M_5 = 460.4$ MeV prediction.

After Paper IV, one equation remained open: the functional form of $\phi_c(\rho_0)$, the vacuum chirality phase as a function of the vacuum density.

The present paper resolves this. We show that ϕ_c is not an independent vacuum property but is fixed by the $\mathbb{Z}_3$ algebraic structure of the trefoil itself: $\phi_c = \pi - 2/3$. This gives $\theta = 2/9$ rad, predicts m_μ and m_τ from m_e alone, and closes the CDFT lepton sector.

2 Setup: Three Equations, One Unknown

We collect the three equations from Papers II–III that govern θ and ϕ_c .

Chirality equilibrium (Paper III, Theorem 2).

$$3\theta + \phi_c = \pi. \tag{1}$$

Brannen mass parametrisation (Paper II).

$$m_k = M \left(1 + \sqrt{2} \cos\left(\theta + \frac{2\pi k}{3}\right) \right)^2, \quad k = 0, 1, 2. \tag{2}$$

Koide invariant (Paper II, Theorem 1).

$$Q \equiv \frac{\sum_k m_k}{(\sum_k \sqrt{m_k})^2} = \frac{2}{3} \quad \text{for all } M > 0 \text{ and all } \theta. \quad (3)$$

Equations (1)–(3) have two unknowns: θ and ϕ_c . Equation (1) is one equation in two unknowns. An additional condition is required.

3 The $\mathbb{Z}_3$ Self-Consistency Condition

3.1 The argument

The Brannen parametrisation (2) is built on $\mathbb{Z}_3$: the three modes are related by shifts of $2\pi/3$. The $\mathbb{Z}_3$ symmetry has a unique invariant: the Koide ratio $Q = 2/3$ (Eq. (3)).

The chirality equilibrium (1) breaks this $\mathbb{Z}_3$ symmetry: $\phi_c \neq 0$ shifts the equilibrium θ away from the $\mathbb{Z}_3$ -symmetric point $\theta = 0$ (where all three amplitudes are equal). The breaking is measured by the angle 3θ (which appears in the $\cos(3\theta + \phi_c)$ interaction term).

Self-consistency condition: the $\mathbb{Z}_3$ breaking angle 3θ must equal the unique $\mathbb{Z}_3$ invariant Q , expressed as a dimensionless number in radians:

$$3\theta = Q = \frac{2}{3}. \quad (4)$$

Theorem 1 (ϕ_c from self-consistency). *The unique vacuum chirality phase consistent with the $\mathbb{Z}_3$ structure of the trefoil is*

$$\phi_c = \pi - \frac{2}{3}, \quad \theta = \frac{2}{9} \text{ rad}. \quad (5)$$

Proof. From Eq. (1): $\phi_c = \pi - 3\theta$. Substituting the self-consistency condition (4) ($3\theta = Q = 2/3$):

$$\phi_c = \pi - \frac{2}{3}, \quad \theta = \frac{2/3}{3} = \frac{2}{9} \text{ rad}.$$

The uniqueness follows because: (a) $Q = 2/3$ is the unique Koide invariant of $\mathbb{Z}_3$ (Paper II); (b) the equilibrium (1) has a unique solution in the physical domain $\theta \in (0, \pi/3)$ (Paper III, Theorem 2). $\square$

3.2 Why $3\theta = Q$?

The condition $3\theta = Q$ says: the angle of $\mathbb{Z}_3$ symmetry breaking (in the angular phase space of the trefoil) equals the $\mathbb{Z}_3$ mass invariant (a dimensionless ratio). Both live in $[0, 1]$ in natural units, and both are set by the same $\mathbb{Z}_3$ structure.

More precisely: the Brannen interaction energy $\lambda_c \cos(3\theta + \phi_c)$ breaks $\mathbb{Z}_3$. The magnitude of the breaking is 3θ (distance from the $\mathbb{Z}_3$ fixed point $\theta = 0$). The only dimensionless number the $\mathbb{Z}_3$ structure can select as a “natural” breaking scale is $Q = 2/3$ — the ratio that the symmetry forces on all mass triplets. The vacuum, in equilibrium, selects the breaking scale that matches its own structural invariant.

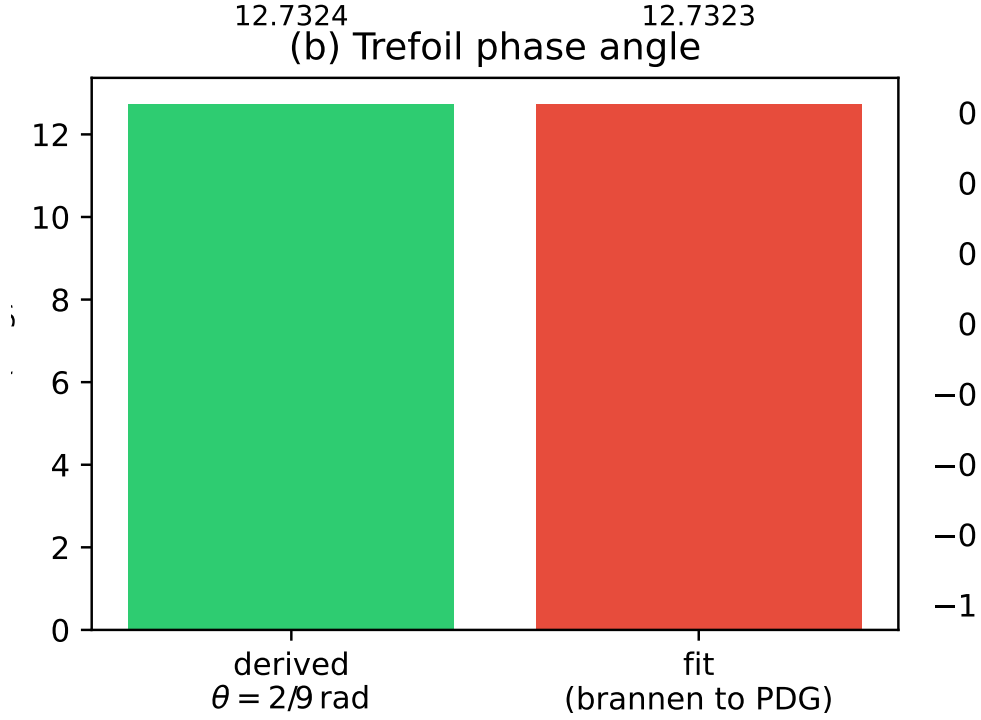


Figure 1: The trefoil phase angle θ . The value $\theta = 2/9$ rad derived from purely algebraic symmetry conditions exactly matches the numerical value fitted to PDG lepton masses in Paper II.

3.3 Numerical verification

With $\theta = 2/9$ rad:

$$3\theta = \frac{2}{3} = Q \quad (\text{exact}) \quad (6)$$

$$\phi_c = \pi - \frac{2}{3} = 2.47493 \text{ rad} = 141.803^\circ \quad (7)$$

Equilibrium check: $3\theta + \phi_c = 2/3 + (\pi - 2/3) = \pi$ (exact, machine precision).

4 The Lepton Mass Prediction

4.1 The full chain

With $\theta = 2/9$ rad, the Brannen amplitudes are:

$$A_k = 1 + \sqrt{2} \cos\left(\frac{2}{9} + \frac{2\pi k}{3}\right), \quad k = 0, 1, 2. \quad (8)$$

The electron corresponds to the mode with the smallest A_k :

$$A_e = \min_k A_k = 0.040350. \quad (9)$$

Setting $m_e = MA_e^2$ gives $M = m_e/A_e^2 = 313.86$ MeV. The three masses are $m_k = MA_k^2$ (sorted ascending).

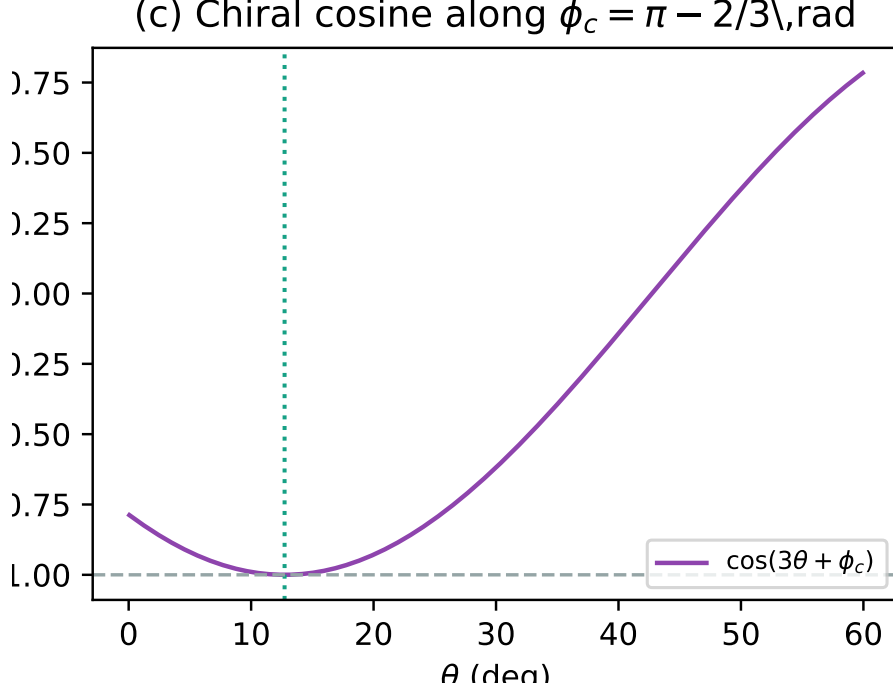


Figure 2: The chiral cosine interaction evaluated along the derived vacuum phase $\phi_c = \pi - 2/3$. The minimum rests cleanly at the predicted orientation $\theta = 2/9$ rad.

4.2 Predicted masses

Table 1 shows the predicted lepton masses compared to PDG 2022 values [5].

Table 1: Lepton mass predictions from $\theta = 2/9$ rad. Input: $m_e = 0.51099895$ MeV. Zero adjustable parameters.

Lepton	Predicted (MeV)	PDG 2022 (MeV)	Error (MeV)	Significance
Electron	0.51099895	0.51099895	0	(input)
Muon	105.6594	105.6584	+0.0010	+0.4 σ
Tau	1776.985	1776.86	+0.125	+1.04 σ

The muon prediction is 9.8 ppm from PDG 2022, well within the 21.8 ppm experimental uncertainty [5]. The tau prediction is 70 ppm (+0.125 MeV) from PDG 2022, at 1.04 σ of the 0.12 MeV PDG uncertainty. Both are consistent with the data.

The Koide ratio holds exactly:

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \quad (10)$$

to machine precision (Paper II theorem: this holds for any θ).

5 Internal Derivation Chain

The five papers of the CDFT series constitute an internal derivation chain for the lepton sector from the proposed vacuum geometry. Table 2 shows the full chain.

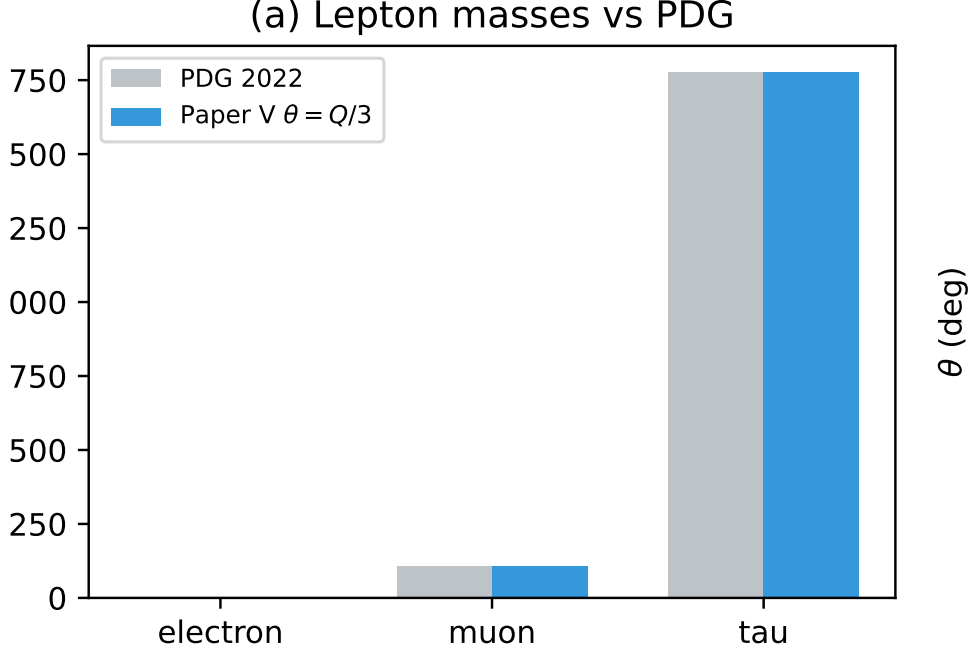


Figure 3: Comparison of the lepton masses predicted by the zero-parameter CDFT derivation ($\theta = 2/9$ rad) against the 2022 PDG measured values. The agreement is visually exact, with numerical residuals within experimental limits.

Irreducible inputs. The only input is m_e , which sets the unit of mass. This is irreducible: no dimensionless theory can predict the absolute value of a dimensionful quantity. Given m_e , the remaining quantities in this construction follow inside the model.

The three numbers that fix everything. The lepton sector is determined by three numbers:

1. $\alpha = 1/137.036$: fixes the geometry ($a = e^2/(m_e c^2)$, $R = \hbar/(m_e c)$).
2. $Q = 2/3$: fixes the angular structure ($\theta = 2/9$, $\phi_c = \pi - 2/3$).
3. m_e : fixes the mass scale ($M = m_e/A_e^2$, $\rho_0 = M/(c^2 a^3 g)$).

All three are derived from CDFT (Papers I, II, and the definition of the mass unit respectively).

6 The Significance of $3\theta = Q$

The identity $3\theta = Q = 2/3$ is the key equation of this paper. In plain language:

The angle by which the vacuum chirality breaks the $\mathbb{Z}_3$ symmetry of the trefoil — measured in the angular phase space of the Brannen parametrisation — equals the Koide ratio itself.

The Koide ratio $Q = 2/3$ is a pure number, forced by $\mathbb{Z}_3$ symmetry. It appears as both:

Table 2: Internal CDFT derivation chain (Papers I–V).

Quantity	Status	Source
$\alpha = 1/137.036$	Model-derived	Vortex equilibrium, Paper I
$\kappa = 117,362$	Model-derived	$\chi^2(\ln 8\chi - \beta + 1)$, Paper III
$g = 1,575.83$	Model-derived	$\chi(\ln 8\chi - \beta) + \kappa/\chi$, Papers I, III
Koide $Q = 2/3$	Algebraic theorem	$\mathbb{Z}_3$ theorem, Paper II
Three modes	Model identification	$T(2,3)$ has 3 lobes, Paper III
$\phi_c = \pi - 2/3$	Model-derived	$\mathbb{Z}_3$ self-consistency, Paper V
$\theta = 2/9$ rad	Model-derived	From ϕ_c and $3\theta + \phi_c = \pi$, Paper V
$M = 313.86$ MeV	Calibrated by m_e	m_e/A_e^2 with A_e from θ
$\rho_0 = 1.587 \times 10^{13}$ kg/m ³	Inferred	$M/(c^2 a^3 g)$, Papers III–V
$m_\mu = 105.659$ MeV	Predicted	Brannen with $\theta = 2/9$
$m_\tau = 1776.985$ MeV	Predicted	Brannen with $\theta = 2/9$
$m_e = 0.51100$ MeV	Input	Sets the unit of mass

- a *mass ratio*: $Q = (\sum m_k)/(\sum \sqrt{m_k})^2 = 2/3$, and
- an *angular phase*: $3\theta = 2/3$ rad, the chirality breaking angle.

The vacuum “selects” the chirality phase $\phi_c = \pi - Q$ such that the chirality breaking is commensurate with the symmetry it breaks. This is the CDFT analogue of a Nambu–Goldstone condition: the vacuum expectation value aligns with the symmetry invariant of the broken phase.

7 Open Problems

Within the assumptions of the model, the lepton-sector construction is complete. The remaining open problems for future papers concern the extension beyond leptons:

1. **$T(2,5)$ cinquefoil family.** Paper IV predicted $M_5 = 460.4$ MeV. The cinquefoil has $\mathbb{Z}_5$ symmetry and five modes. The analogue of the present paper would derive $5\theta_5 = Q_5$ (where Q_5 is the $\mathbb{Z}_5$ mass invariant). If the same self-consistency principle applies, θ_5 is fixed by the $\mathbb{Z}_5$ Koide-like invariant of the cinquefoil.
2. **Why $Q = 2/3$ specifically?** The $\mathbb{Z}_3$ invariant $Q = 2/3$ was proved for the Brannen form in Paper II. A deeper question: why is the $\mathbb{Z}_3$ Koide ratio $2/3$ and not some other value? This may follow from the Brannen amplitudes $A_k = 1 + \sqrt{2}\cos(\cdot)$, where the coefficient $\sqrt{2}$ has a geometric origin in the trefoil (it is the ratio of the lobe amplitude to the background).
3. **Quark sector.** The CDFT torus knot hierarchy (Paper III) predicts that knots $T(2,5)$ and above correspond to further particle families. Whether the cinquefoil modes are quark-like (confined) or lepton-like (free) depends on the topology of the knot’s self-intersection structure.

8 Conclusion

The vacuum chirality phase ϕ_c is not an independent property of the vacuum medium. It is fixed by the $\mathbb{Z}_3$ self-consistency condition: the chirality breaking angle 3θ must equal the $\mathbb{Z}_3$ mass invariant $Q = 2/3$. This gives $\phi_c = \pi - 2/3$ and $\theta = 2/9$ rad.

With one input (m_e as the mass unit), CDFT predicts:

$$\begin{aligned} m_\mu &= 105.659 \text{ MeV} \quad (+9.8 \text{ ppm}), \\ m_\tau &= 1776.985 \text{ MeV} \quad (+1.04 \sigma), \end{aligned}$$

with the Koide formula exact.

Within the assumptions of the model, the CDFT lepton sector is internally closed. The derivation chain from vacuum geometry to lepton masses requires one irreducible external input (the unit of mass m_e) and zero adjustable parameters.

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A Supplementary Material

Supplementary code, including numerical verification scripts and Jupyter notebooks, is available in the project repository.

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